

Docker CIS Compliance Report

Generated on 2026-04-29 15:29:42

Compliance Score	58%
Checks Passed	31 / 53
Status	PASS: 31 FAIL: 22 WARN: 4 N/A: 11
Failures by Severity	CRITICAL: 4 HIGH: 9 MEDIUM: 4 LOW: 5

Host Configuration (0/2 passed)

Control	Severity	Status	Description	Details
1.2.1	HIGH	ERROR	Ensure auditing is configured for the Docker daemon	Command '['auditpol', '/get', '/category:Detailed Tracking']' returned non-zero exit status 1314.
1.1.2	HIGH	MANUAL_REVIEW	Ensure only trusted users are allowed to control Docker daemon	Users in docker group: HP

Daemon Configuration (1/4 passed)

Control	Severity	Status	Description	Details
2.4	HIGH	FAIL	Ensure insecure registries are not used	Insecure registries configured: ::1/128
Daemon_Status	CRITICAL	PASS	Ensure Docker daemon is running	Docker daemon is active and reachable
2.11	MEDIUM	FAIL	Ensure centralized and remote logging is configured	Current logging driver: json-file
2.6	CRITICAL	MANUAL_REVIEW	Ensure TLS authentication for Docker daemon is configured	TLS flags not in process args. Check daemon.json manually.

Remediation for 2.4:
Remove insecure registries from /etc/docker/daemon.json:

Remove the "insecure-registries" key or set it to an empty list:

```
{"insecure-registries": []}
```

Then restart Docker: `sudo systemctl restart docker`

Ref: CIS Docker Benchmark v1.6.0, Section 2.4

Remediation for 2.11:

Configure a remote logging driver in `/etc/docker/daemon.json`:

```
{"log-driver": "syslog", "log-opts": {"syslog-address": "tcp://your-log-server:514"}}
```

Other options: splunk, gelf, fluentd, awslogs

Then restart Docker: `sudo systemctl restart docker`

Ref: CIS Docker Benchmark v1.6.0, Section 2.11

Socket Configuration (0/4 passed)

Control	Severity	Status	Description	Details
3.15	HIGH	N/A	Ensure that the Docker socket file ownership is set to root:docker	Socket not found
3.16	HIGH	N/A	Ensure that the Docker socket file permissions are set to 660 or more restrictively	Socket not found
3.3	MEDIUM	N/A	Ensure that docker.socket file ownership is set to root:root	File not found
3.4	MEDIUM	N/A	Ensure that docker.socket file permissions are set to 644 or more restrictive	File not found

Container Runtime (0/7 passed)

Control	Severity	Status	Description	Details
5.32	CRITICAL	FAIL	Ensure that the Docker socket is not mounted inside any containers	Containers mounting docker.sock: cis_test_vulnerable
5.10	CRITICAL	FAIL	Ensure that the host's network namespace is not shared	Containers using host network: cis_test_vulnerable
5.11	HIGH	FAIL	Ensure that the memory usage for containers is limited	Containers without memory limits: cis_test_vulnerable
5.5	CRITICAL	FAIL	Ensure that privileged containers are not used	Privileged containers: cis_test_vulnerable
4.1	HIGH	FAIL	Ensure that a user for the container has been created (Non-root)	Running as root: cis_test_vulnerable
5.13	MEDIUM	FAIL	Ensure that the container's root filesystem is mounted as read only	Containers with writable rootfs: cis_test_vulnerable
5.6	CRITICAL	FAIL	Ensure sensitive host system directories are not mounted	Violations: {'cis_test_vulnerable': ['/']}

Remediation for 5.32:

Do NOT mount the Docker socket inside containers:

Remove `-v /var/run/docker.sock:/var/run/docker.sock` from your docker run command.

If you need Docker API access, use a restricted proxy like `docker-socket-proxy`:

```
docker run -v /var/run/docker.sock:/var/run/docker.sock tecnativa/docker-socket-proxy
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.32

Remediation for 5.10:

Do NOT use `--network=host`. Use bridge (default) or custom networks:

```
docker run --network=bridge
```

Or create a custom network:

```
docker network create mynet
```

```
docker run --network=mynet
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.10

Remediation for 5.11:

Set memory limits when running containers:

```
docker run --memory=512m --memory-swap=1g
```

In `docker-compose.yml`:

```
deploy:
```

```
resources:
```

```
limits:
```

```
memory: 512M
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.11

Remediation for 5.5:

Do NOT use the `--privileged` flag. Instead, grant only specific capabilities:

```
docker run --cap-drop ALL --cap-add NET_BIND_SERVICE
```

If a container currently uses `--privileged`, re-create it without that flag:

```
docker stop && docker rm
```

```
docker run --cap-drop ALL --cap-add
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.5

Remediation for 4.1:

Always specify a non-root user in your Dockerfile:

```
RUN useradd -r appuser
```

```
USER appuser
```

Or pass `--user` at runtime:

```
docker run --user 1000:1000
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.1

Remediation for 5.13:

Run containers with a read-only root filesystem:

```
docker run --read-only
```

Use tmpfs for writable directories:

```
docker run --read-only --tmpfs /tmp --tmpfs /run
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.13

Remediation for 5.6:

Do NOT mount sensitive host directories. Remove volume mounts for:

```
/, /boot, /dev, /etc, /lib, /proc, /sys, /usr
```

Use named volumes instead of host bind mounts:

```
docker volume create app-data
```

```
docker run -v app-data:/app/data
```

Ref: CIS Docker Benchmark v1.6.0, Section 5.6

Image Security (2/6 passed)

Control	Severity	Status	Description	Details
IMG-03	HIGH	FAIL	Ensure Content Trust for Docker is enabled	DOCKER_CONTENT_TRUST is not set. Image pulls are not signature-verified.
IMG-04	LOW	PASS	Ensure there are no dangling images	No dangling images found.
IMG-05	LOW	PASS	Ensure Docker images are not excessively large	All images are under 1.00 GB.
IMG-01	MEDIUM	FAIL	Ensure images are not tagged with 'latest'	Images using latest/untagged: test_env-vulnerable-app:latest, test_env-secure-app:latest, project-3_team_34-service:latest, postgres:latest, nginx:latest, alpine:latest
IMG-02	MEDIUM	FAIL	Ensure Docker images are not outdated	Stale images (>90d): alpine:latest (91d old)
IMG-06	CRITICAL	N/A	Ensure Docker images are scanned for vulnerabilities	Scanner execution failed for 'test_env-vulnerable-app:latest'. Ensure Docker Scout or Trivy is properly configured.

Remediation for IMG-03:

Enable Docker Content Trust:

```
export DOCKER_CONTENT_TRUST=1
```

To make it permanent, add to your shell profile (~/.bashrc or ~/.profile):

```
echo 'export DOCKER_CONTENT_TRUST=1' >> ~/.bashrc
```

Windows PowerShell:

```
[System.Environment]::SetEnvironmentVariable('DOCKER_CONTENT_TRUST','1','User')
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.5

Remediation for IMG-01:

Always use specific version tags instead of :latest:

```
docker pull nginx:1.25.3 (instead of docker pull nginx:latest)
```

In Dockerfile:

```
FROM python:3.12-slim (instead of FROM python:latest)
```

Re-tag existing images:

```
docker tag myapp:latest myapp:1.0.0
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.7

Remediation for IMG-02:

Rebuild images from updated base images regularly:

```
docker pull :
```

```
docker build --no-cache -t myapp:latest .
```

Set up a CI/CD pipeline to rebuild images on a schedule (e.g., weekly).

Ref: Docker Security Best Practices – Image Freshness

Dockerfile Security (28/48 passed)

Control	Severity	Status	Description	Details
DL-03	LOW	WARN	Ensure ADD is not used for remote URLs	[Image 3a44c13c4be3] ADD instructions found (use COPY for local files, curl/wget for remote): Line 15: ADD used instead of COPY
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image 3a44c13c4be3] All apt-get install commands use --no-install-recommends.

Control	Severity	Status	Description	Details
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image 3a44c13c4be3] Port 22 is not exposed.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image 3a44c13c4be3] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.
DL-06	LOW	FAIL	Ensure HEALTHCHECK instructions have been added to container images	[Image 3a44c13c4be3] No HEALTHCHECK instruction found.
DL-02	HIGH	FAIL	Ensure a non-root USER is specified	[Image 3a44c13c4be3] No USER instruction found. Container will run as root by default.
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image 3a44c13c4be3] No ENV instructions found.
DL-07	MEDIUM	PASS	Ensure update instructions are not used alone	[Image 3a44c13c4be3] No isolated update instructions found.
DL-03	LOW	PASS	Ensure ADD is not used for remote URLs	[Image bba7ab5676b6] No ADD instructions found. COPY is used correctly.
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image bba7ab5676b6] No apt-get install commands found.
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image bba7ab5676b6] No EXPOSE instructions found.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image bba7ab5676b6] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.
DL-06	LOW	PASS	Ensure HEALTHCHECK instructions have been added to container images	[Image bba7ab5676b6] HEALTHCHECK instruction found (line 6).
DL-02	HIGH	PASS	Ensure a non-root USER is specified	[Image bba7ab5676b6] Non-root USER 'appuser' is set (line 8).
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image bba7ab5676b6] No ENV instructions found.
DL-07	MEDIUM	PASS	Ensure update instructions are not used alone	[Image bba7ab5676b6] No isolated update instructions found.
DL-03	LOW	WARN	Ensure ADD is not used for remote URLs	[Image 25109184c71b] ADD instructions found (use COPY for local files, curl/wget for remote): Line 1: ADD used instead of COPY
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image 25109184c71b] No RUN instructions found.
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image 25109184c71b] No EXPOSE instructions found.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image 25109184c71b] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.

Control	Severity	Status	Description	Details
DL-06	LOW	FAIL	Ensure HEALTHCHECK instructions have been added to container images	[Image 25109184c71b] No HEALTHCHECK instruction found.
DL-02	HIGH	FAIL	Ensure a non-root USER is specified	[Image 25109184c71b] No USER instruction found. Container will run as root by default.
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image 25109184c71b] No ENV instructions found.
DL-07	MEDIUM	PASS	Ensure update instructions are not used alone	[Image 25109184c71b] No RUN instructions found.
DL-03	LOW	WARN	Ensure ADD is not used for remote URLs	[Image 55ecfa64be1c] ADD instructions found (use COPY for local files, curl/wget for remote): Line 3: ADD used instead of COPY
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image 55ecfa64be1c] No apt-get install commands found.
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image 55ecfa64be1c] Port 22 is not exposed.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image 55ecfa64be1c] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.
DL-06	LOW	FAIL	Ensure HEALTHCHECK instructions have been added to container images	[Image 55ecfa64be1c] No HEALTHCHECK instruction found.
DL-02	HIGH	FAIL	Ensure a non-root USER is specified	[Image 55ecfa64be1c] No USER instruction found. Container will run as root by default.
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image 55ecfa64be1c] No secrets found in ENV instructions.
DL-07	MEDIUM	WARN	Ensure update instructions are not used alone	[Image 55ecfa64be1c] Isolated update commands found: Line 5: apk update used without add/upgrade
DL-03	LOW	PASS	Ensure ADD is not used for remote URLs	[Image 6e23479198b9] No ADD instructions found. COPY is used correctly.
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image 6e23479198b9] All apt-get install commands use --no-install-recommends.
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image 6e23479198b9] Port 22 is not exposed.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image 6e23479198b9] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.
DL-06	LOW	FAIL	Ensure HEALTHCHECK instructions have been added to container images	[Image 6e23479198b9] No HEALTHCHECK instruction found.
DL-02	HIGH	FAIL	Ensure a non-root USER is specified	[Image 6e23479198b9] No USER instruction found. Container will run as root by default.

Control	Severity	Status	Description	Details
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image 6e23479198b9] No secrets found in ENV instructions.
DL-07	MEDIUM	PASS	Ensure update instructions are not used alone	[Image 6e23479198b9] No isolated update instructions found.
DL-03	LOW	PASS	Ensure ADD is not used for remote URLs	[Image 78481659c47e] No ADD instructions found. COPY is used correctly.
DL-05	LOW	PASS	Ensure apt-get uses --no-install-recommends	[Image 78481659c47e] All apt-get install commands use --no-install-recommends.
DL-04	HIGH	PASS	Ensure SSH port 22 is not exposed	[Image 78481659c47e] Port 22 is not exposed.
DL-01	MEDIUM	N/A	Ensure FROM does not use :latest tag	[Image 78481659c47e] FROM instructions are not preserved in image history. See Image Security checks (IMG-01) for tag validation.
DL-06	LOW	FAIL	Ensure HEALTHCHECK instructions have been added to container images	[Image 78481659c47e] No HEALTHCHECK instruction found.
DL-02	HIGH	FAIL	Ensure a non-root USER is specified	[Image 78481659c47e] No USER instruction found. Container will run as root by default.
DL-08	CRITICAL	PASS	Ensure secrets are not stored in Dockerfile ENV instructions	[Image 78481659c47e] No secrets found in ENV instructions.
DL-07	MEDIUM	PASS	Ensure update instructions are not used alone	[Image 78481659c47e] No isolated update instructions found.

Remediation for DL-03:

Use COPY instead of ADD unless you specifically need to extract a tar file.
 For downloading remote files, use curl or wget inside a RUN instruction:
 RUN curl -fsSL https://example.com/file.tar.gz | tar -xz
 Ref: CIS Docker Benchmark v1.6.0, Section 4.9

Remediation for DL-06:

Add a HEALTHCHECK instruction to your Dockerfile:
 HEALTHCHECK --interval=5m --timeout=3s \\
 CMD curl -f http://localhost/ || exit 1
 Ref: CIS Docker Benchmark v1.6.0, Section 4.6

Remediation for DL-02:

Add a USER instruction to your Dockerfile to run as a non-root user:
 RUN groupadd -r appuser && useradd -r -g appuser appuser
 USER appuser
 Ref: CIS Docker Benchmark v1.6.0, Section 4.1

Remediation for DL-03:

Use COPY instead of ADD unless you specifically need to extract a tar file.
 For downloading remote files, use curl or wget inside a RUN instruction:
 RUN curl -fsSL https://example.com/file.tar.gz | tar -xz
 Ref: CIS Docker Benchmark v1.6.0, Section 4.9

Remediation for DL-06:

Add a HEALTHCHECK instruction to your Dockerfile:
 HEALTHCHECK --interval=5m --timeout=3s \\
 CMD curl -f http://localhost/ || exit 1
 Ref: CIS Docker Benchmark v1.6.0, Section 4.6

Remediation for DL-02:

Add a USER instruction to your Dockerfile to run as a non-root user:

```
RUN groupadd -r appuser && useradd -r -g appuser appuser
USER appuser
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.1

Remediation for DL-03:

Use COPY instead of ADD unless you specifically need to extract a tar file.

For downloading remote files, use curl or wget inside a RUN instruction:

```
RUN curl -fsSL https://example.com/file.tar.gz | tar -xz
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.9

Remediation for DL-06:

Add a HEALTHCHECK instruction to your Dockerfile:

```
HEALTHCHECK --interval=5m --timeout=3s \
  CMD curl -f http://localhost/ || exit 1
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.6

Remediation for DL-02:

Add a USER instruction to your Dockerfile to run as a non-root user:

```
RUN groupadd -r appuser && useradd -r -g appuser appuser
USER appuser
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.1

Remediation for DL-07:

Combine apt-get update and apt-get install into a single RUN instruction:

```
RUN apt-get update && apt-get install -y
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.7

Remediation for DL-06:

Add a HEALTHCHECK instruction to your Dockerfile:

```
HEALTHCHECK --interval=5m --timeout=3s \
  CMD curl -f http://localhost/ || exit 1
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.6

Remediation for DL-02:

Add a USER instruction to your Dockerfile to run as a non-root user:

```
RUN groupadd -r appuser && useradd -r -g appuser appuser
USER appuser
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.1

Remediation for DL-06:

Add a HEALTHCHECK instruction to your Dockerfile:

```
HEALTHCHECK --interval=5m --timeout=3s \
  CMD curl -f http://localhost/ || exit 1
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.6

Remediation for DL-02:

Add a USER instruction to your Dockerfile to run as a non-root user:

```
RUN groupadd -r appuser && useradd -r -g appuser appuser
USER appuser
```

Ref: CIS Docker Benchmark v1.6.0, Section 4.1